VULNERABILITY ASSESMENT REPORT

PROJECT 2

Vulnerability Assessment of Windows Operating System Using Tenable Nessus Essentials

Prepared by:

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14 February 2026

DECLARATION

I hereby declare that the project report titled **"Vulnerability Assessment of Windows Operating System Using Tenable Nessus Essentials"** is my original work carried out by me as part of the academic requirements for the subject **Cyber Security**.

I confirm that this report has not been submitted previously to any other institution or university for the award of any degree or certification.

All the information presented in this report is based on the vulnerability assessment performed on my system using Tenable Nessus Essentials, and the findings, analysis, remediation, and conclusions are documented truthfully to the best of my knowledge.

I take full responsibility for the authenticity and accuracy of the work presented in this report.

Date: 15 February 2026

Place: Visakhapatnam

Signature: Ch. Ujwala

Name: Chippada Ujwala

ABSTRACT

This report presents a vulnerability assessment conducted on a Windows Operating System using Tenable Nessus Essentials. The objective of this assessment was to identify security weaknesses, misconfigurations, and potential threats that could be exploited by attackers.

The scanning process was performed on the target system within a controlled environment. Nessus Essentials was used to detect vulnerabilities based on known security flaws, configuration issues, and exposed services. The assessment identified multiple informational findings and one medium-severity vulnerability related to SMB configuration.

The identified vulnerabilities were analyzed, classified based on severity levels, and prioritized according to risk impact. Remediation steps were implemented where possible to enhance the security posture of the system.

This assessment demonstrates the importance of vulnerability scanning, risk prioritization, and system hardening in maintaining a secure computing environment.

INTRODUCTION

In today's digital era, operating systems are constantly exposed to security threats such as malware, unauthorized access, and network-based attacks. A vulnerability assessment is a systematic process used to identify, analyze, and prioritize security weaknesses in a system before they can be exploited by attackers.

Windows Operating Systems are widely used in academic, business, and personal environments. Due to their popularity, they are frequent targets for cyber threats. Therefore, it is essential to regularly assess and strengthen their security configurations.

This project focuses on performing a vulnerability assessment on a Windows Operating System using Tenable Nessus Essentials. Nessus is a widely recognized vulnerability scanning tool that detects misconfigurations, outdated software, insecure protocols, and other potential security risks.

The main objective of this assessment is to:

*Identify vulnerabilities present in the system

*Classify and prioritize risks based on severity

*Analyze potential business impact

*Provide remediation and hardening recommendations

Through this assessment, the importance of proactive security monitoring and system hardening is demonstrated.

OBJECTIVES OF THE PROJECT

The primary objectives of this project are:

1. To perform a vulnerability assessment on a Windows Operating System.
2. To identify security weaknesses using Tenable Nessus Essentials.
3. To classify vulnerabilities based on severity levels (Critical, High, Medium, Low, Informational).
4. To prioritize risks based on impact and exploitability.
5. To implement remediation measures for identified vulnerabilities.
6. To validate remediation through a follow-up scan.
7. To recommend system hardening strategies for improved security.

TOOLS USED

1. Tenable Nessus Essentials

Tenable Nessus Essentials is a widely used vulnerability assessment tool designed to detect security weaknesses, configuration issues, and known vulnerabilities in operating systems and network services.

It scans systems using a database of known vulnerabilities and classifies findings based on severity levels using the Common Vulnerability Scoring System (CVSS).

Tool Details:

- Tool Name: Tenable Nessus Essentials
- Tool Type: Vulnerability Scanner
- Scanner Mode: Local Scanner
- Scan Policy Used: Basic Network Scan
- Severity Classification Standard: CVSS v3.0

Nessus Essentials was used in this project to perform a non-credentialed scan on the Windows Operating System and identify potential security risks.

TARGET SYSTEM DETAILS

The vulnerability assessment was conducted on the following target system:

- Target Operating System: Windows 11
- Target IP Address: 192.168.0.105
- Network Environment: Local Network
- Scanner Type: Local Scanner
- Scan Type: Non-Credentialed Basic Network Scan
- Scan Date: 14 February 2026

The scan was performed within a controlled environment. The Windows Firewall was temporarily disabled during the assessment as per project instructions and re-enabled after completion of the scan.

The objective of targeting this system was to identify configuration weaknesses, exposed services, and protocol-level vulnerabilities that could pose potential security risks.

METHODOLOGY

The vulnerability assessment was conducted using the following structured approach:

1. **Installation and Configuration**
   Tenable Nessus Essentials was installed and configured on the local system.
2. **Identification of Target**
   The target system IP address was identified using the ipconfig command.
3. **Pre-Scan Preparation**
   Windows Defender Firewall was temporarily disabled as per project requirements to allow complete scanning.
4. **Scan Configuration**
   A Basic Network Scan was configured in Nessus by entering the target IP address.
5. **Scan Execution**
   The vulnerability scan was launched and allowed to complete. Nessus analyzed open ports, services, and configurations against known vulnerability databases.
6. **Analysis of Results**
   The detected vulnerabilities were reviewed and classified based on severity levels.
7. **Risk Prioritization**
   Medium severity vulnerabilities were prioritized for remediation.
8. **Remediation Implementation**
   SMB signing policies were enabled to mitigate the identified medium severity vulnerability.
9. **Validation Scan**
   A follow-up scan was performed after system restart to verify remediation effectiveness.

SCAN RESULTS SUMMARY

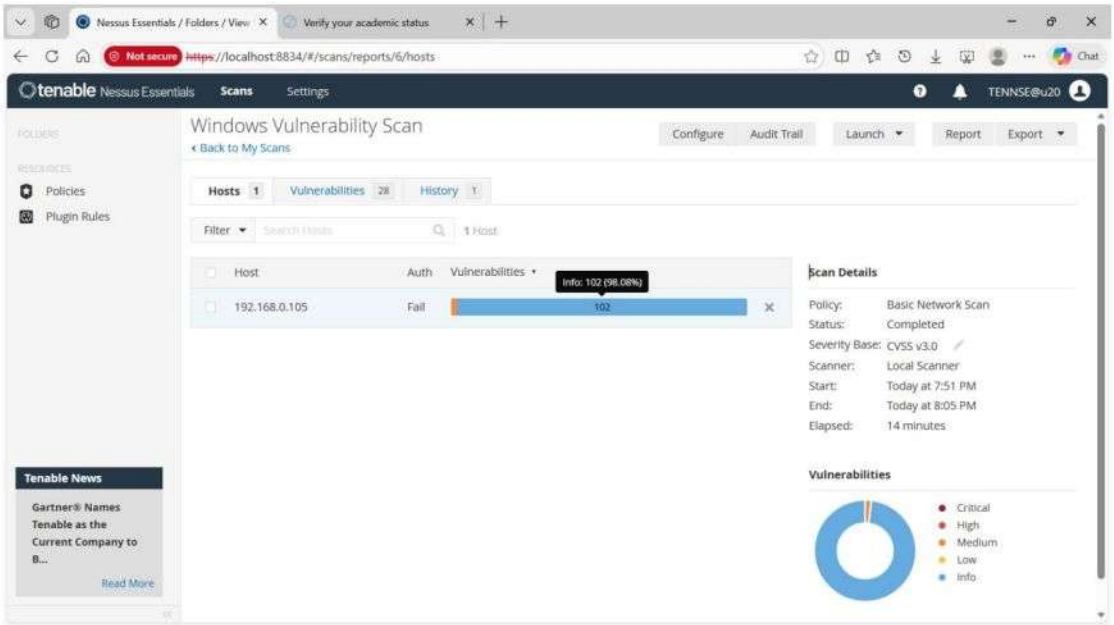
The initial vulnerability scan identified multiple security findings on the target Windows system.

Summary of Findings:

- Total Vulnerabilities Detected: 102
- Medium Severity: 2
- Informational: Majority
- High Severity: 0
- Critical Severity: 0

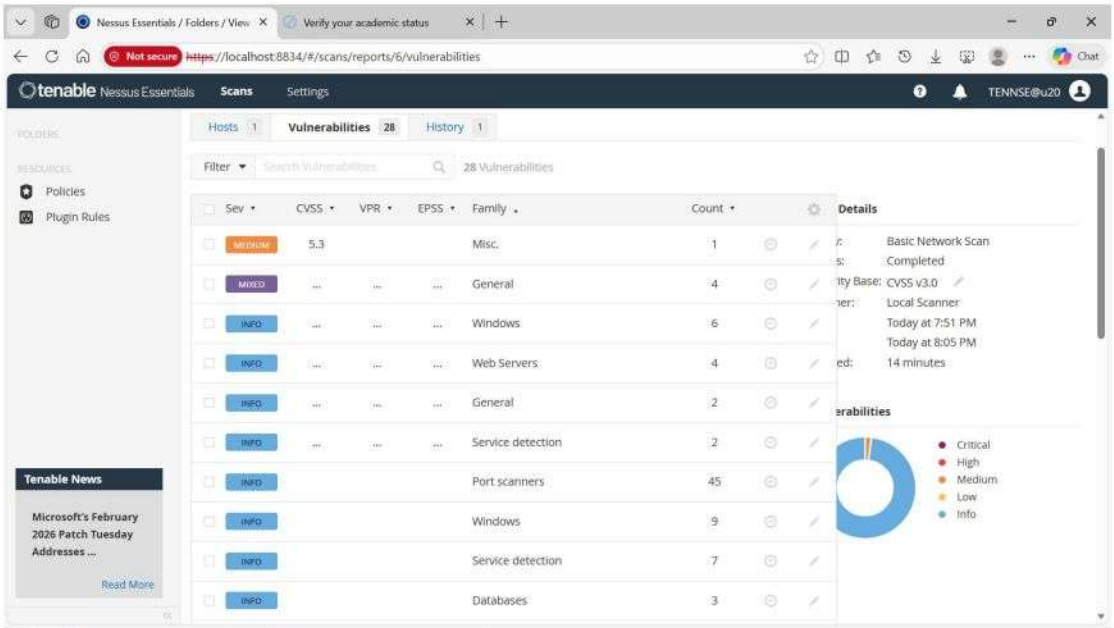
The majority of the findings were informational in nature, indicating configuration details and exposed services. One medium severity vulnerability related to SMB configuration was identified and prioritized for remediation.

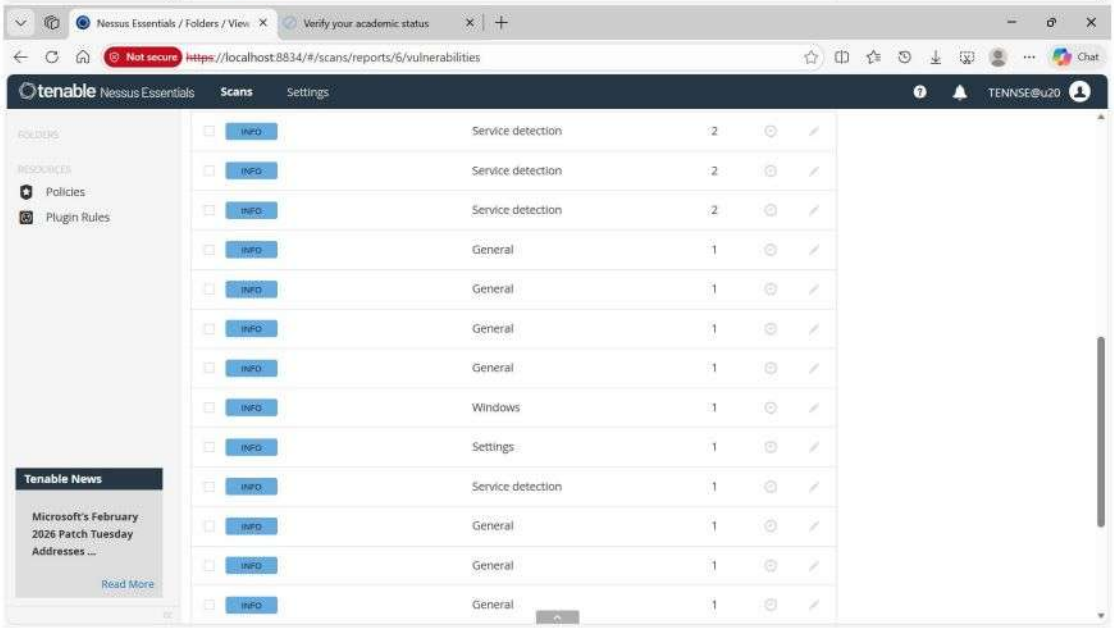
The scan duration was approximately 14 minutes, and the assessment was conducted using CVSS v3.0 severity classification.

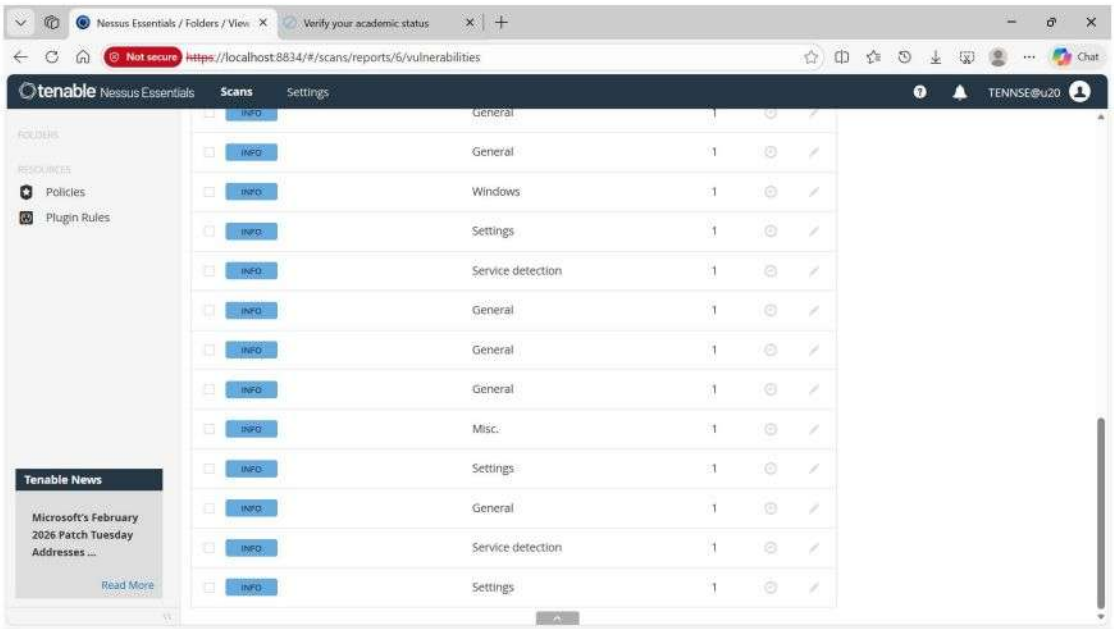


VULNERABILITES IDENTIFIED

The following screenshots represent the vulnerabilities detected during the scan.






Nessus Essentials / Folders / View
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Read More
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INFO Misc. 1
INFO Settings 1
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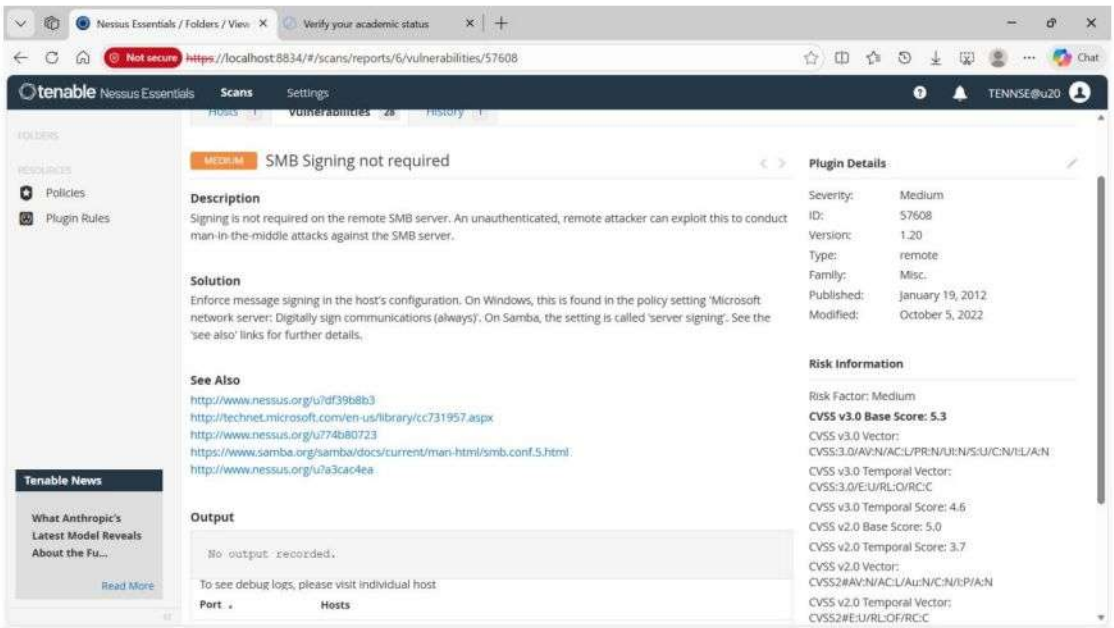
MEDIUM SEVERITY VULNERABILITY

Vulnerability Name: SMB Signing Not Required

Plugin ID: 57608

Severity: Medium

CVSS v3.0 Base Score: 5.3



Description:

The scan identified that SMB (Server Message Block) signing is not required on the target Windows system. When SMB signing is not enforced, an attacker can potentially perform a Man-in-the-Middle (MITM) attack by intercepting and modifying communication between the client and the server.

Risk Impact:

- Possibility of credential interception
- Data tampering during network communication
- Increased risk of session hijacking
- Exposure of internal network communication

Although this vulnerability is rated as Medium, it can be leveraged in combination with other weaknesses to perform more serious attacks.

Remediation:

To fix this issue, the following configuration was implemented:

1. Open **Local Security Policy**
2. Navigate to:
   Local Policies → Security Options
3. Enable:
   Microsoft network server: Digitally sign communications (always)
4. Restart the system to apply changes

This ensures secure SMB communication and prevents unauthorized manipulation of network traffic.

INFORMATIONAL VULNERABILITIES

The majority of detected vulnerabilities were informational in nature. These findings do not directly represent exploitable security flaws but provide useful system information that could assist an attacker in reconnaissance.

Examples Identified:

- Open Port Detection (TCP ports such as 445, 135)
- SMB Service Information Disclosure
- Windows Version Disclosure
- NetBIOS Name Disclosure
- Service Detection Findings
- Plugin Output Information

Risk Impact:

While informational findings are not directly harmful, they may:

- Help attackers map the network
- Reveal system version details
- Assist in targeted exploit attempts
- Increase exposure to reconnaissance-based attacks

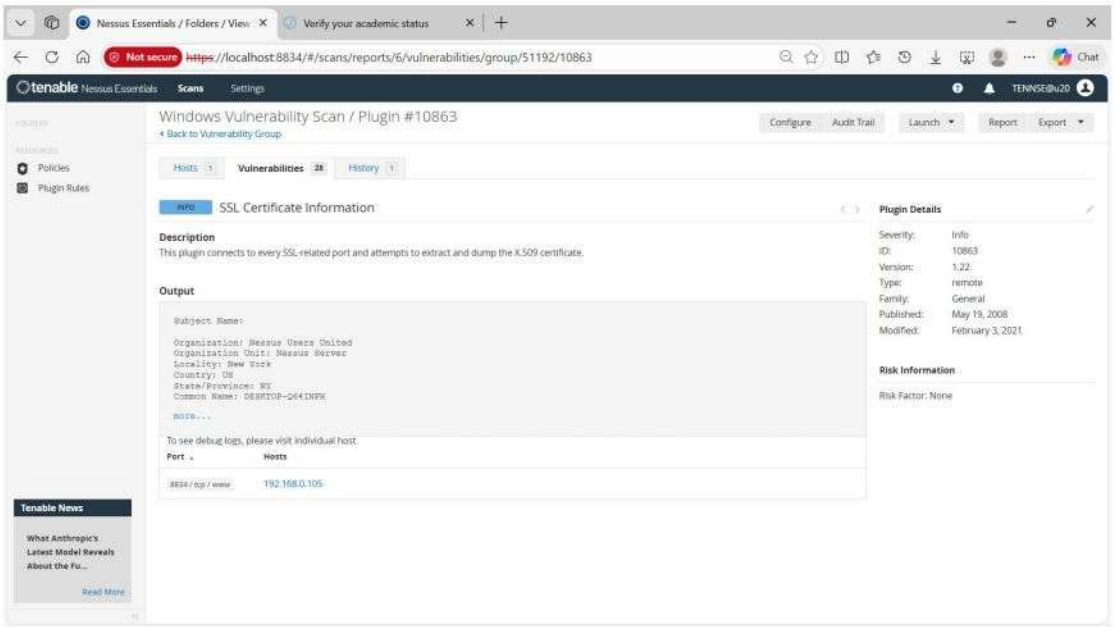
Recommended Hardening Measures:

- Disable unnecessary services
- Close unused ports
- Enable firewall rules
- Limit SMB exposure
- Restrict anonymous accesS
- Keep system updated with latest patches

VULNERABILITY 2

SSL Certificate Information Disclosure

Severity: Informational
Plugin ID: 10863
Port: 8834/tcp



Description

The Nessus scanner identified SSL certificate information exposed on the target system. The plugin connects to SSL-related ports and extracts X.509 certificate details.

The scan revealed:

- Organization Name
- Organization Unit
- Location
- Country
- Common Name (System Name)

Risk Impact

This is an informational finding and does not represent a direct vulnerability.

However, exposed certificate details may:

- Reveal internal system naming conventions
- Disclose organizational structure
- Assist attackers during reconnaissance
- Help in crafting targeted phishing or social engineering attacks

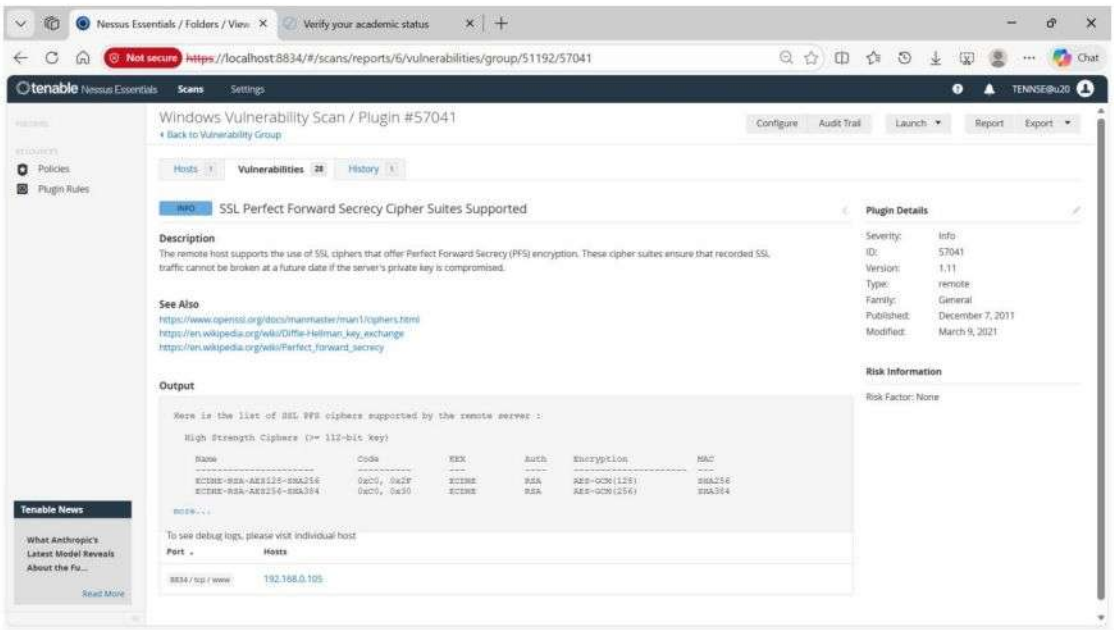
Recommended Mitigation

- Ensure certificates do not reveal unnecessary internal details
- Use properly configured and trusted SSL certificates
- Avoid exposing internal hostnames in publicly accessible services
- Keep SSL/TLS configuration hardened

VULNERABILITY 3

SSL Perfect Forward Secrecy (PFS) Cipher Suites Supported

Severity: Informational
Plugin ID: 57041
Port: 8834/tcp



Description

The Nessus scan detected that the target system supports SSL cipher suites that provide **Perfect Forward Secrecy (PFS)**.

Perfect Forward Secrecy ensures that even if the server's private key is compromised in the future, previously recorded encrypted sessions cannot be decrypted.

The scan identified high-strength cipher suites such as:

- ✧ ECDHE-RSA-AES128-SHA256
- ✧ ECDHE-RSA-AES256-SHA384

Security Impact

This is a **positive security configuration**.

It indicates:

- Strong encryption algorithms are enabled
- Secure key exchange mechanism (ECDHE) is in use
- Protection against future key compromise

This reduces the risk of data exposure from intercepted encrypted traffic.

Recommendation

No remediation required.

Continue:

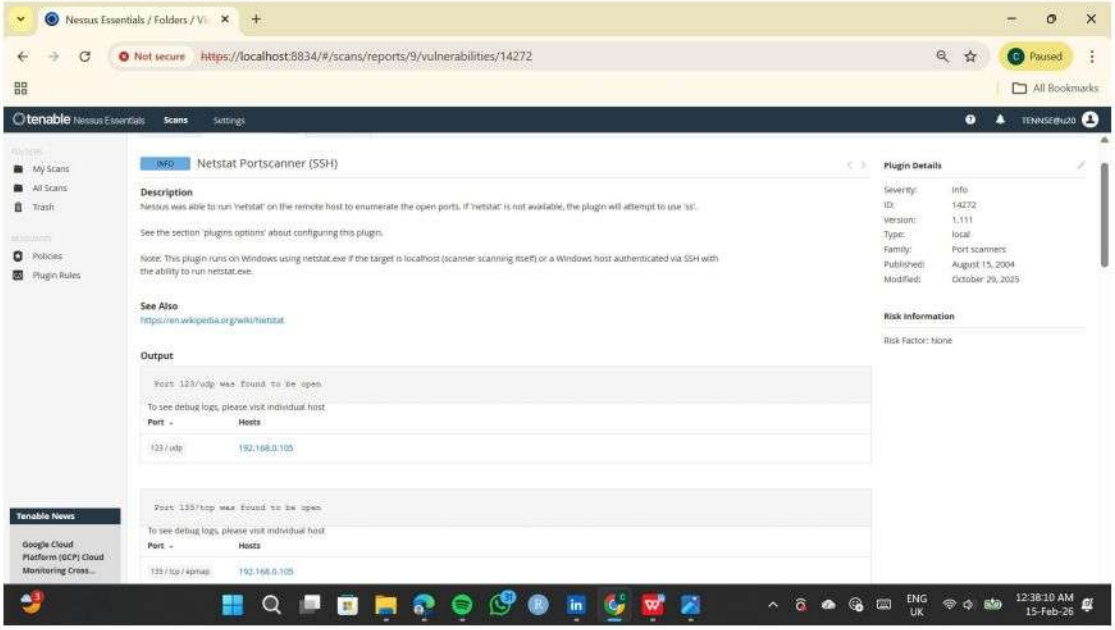
- Using strong TLS configurations
- Disabling weak/legacy cipher suites
- Keeping SSL/TLS updated

VULNERABILITY 4

Netstat Portscanner (ssh) – Informational

Severity: Informational
Plugin ID: 14272

Description:
The Nessus scanner used the netstat utility to enumerate open ports on the target Windows system. The scan revealed that ports 123/UDP and 135/TCP were open.



Impact:
Open ports may expose services to the network. Port 135 (RPC Endpoint Mapper) is commonly targeted for enumeration and potential exploitation if misconfigured.

Recommendation:

- Disable unused services.
- Restrict port access using Windows Firewall rules.
- Monitor exposed services regularly.

VULNERABILITY 5

Smartbedded Meteobridge Web Detection

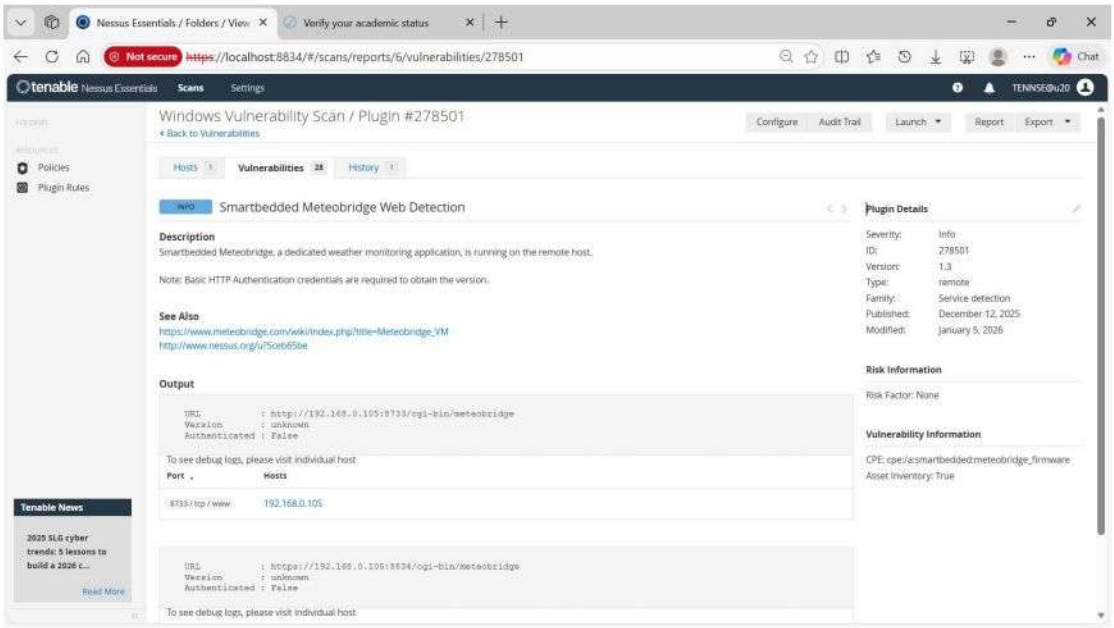
Plugin ID: 278501
Severity: Informational
Port: 8733 / 8834 (TCP)

Description:

The scan detected a Smartbedded Meteobridge web application running on the remote host. The service was accessible through HTTP/HTTPS interfaces.

Technical Details:

- URL: http://192.168.0.105:8733/cgi-bin/meteobridge
- URL: https://192.168.0.105:8834/cgi-bin/meteobridge
- Authentication: Not enabled
- Version: Unknown



Risk Impact:

Although categorized as informational, exposed web services may provide system information to attackers and increase the attack surface of the system.

Recommendation:

- Disable unnecessary services.
- Restrict access using firewall rules.
- Enable authentication for web interfaces.
- Limit exposure to trusted internal networks only.

REMAINING VULNERABILITIES SUMMARY

S.No	Plugin Name	Plugin ID	Severity	Port	Description (Short)
1	SSL Certificate Information	10863	Info	8834	SSL certificate details exposed
2	SSL Perfect Forward Secrecy Cipher Suites Supported	57041	Info	8834	Server supports PFS cipher suites
3	Microsoft Windows SMB Service Detection	51192	Info	445	SMB service detected
4	Microsoft Windows SMB NativeLanMan	—	Info	445	SMB NativeLanMan information disclosure
5	Microsoft Windows SMB Versions Supported	—	Info	445	SMB protocol versions identified
6	Microsoft Windows SMB2 and SMB3 Dialects Supported	—	Info	445	SMB2/SMB3 dialect enumeration
7	Windows NetBIOS / SMB Remote Host Information Disclosure	—	Info	137/139	NetBIOS information exposed
8	Netstat Portscanner (SSH)	14272	Info	135/137	Open ports detected
9	DCE Services Enumeration	10736	Info	135	DCE/RPC services enumerated
10	SMB signing not required	57608	Medium	445	SMB signing disabled (mitm risk)

Risk Prioritization

Then write:

- The vulnerabilities were categorized based on CVSS v3 scoring system provided by Nessus.
- Medium Risk: 1 vulnerability (SMB Signing Not Required – CVSS 5.3)
- Informational: 27 vulnerabilities
- High Risk: None
- Critical Risk: None

Priority Order:

1. Medium Severity Vulnerability
2. Service Exposure Issues
3. Informational Findings

REMEDIATION FOR VULNERBILITY (MIXED)

Vulnerability 1: SMB Signing Not Required

Plugin ID: 57608
Severity: Medium
CVSS v3 Score: 5.3

Vulnerability Explanation

SMB (Server Message Block) signing is not enforced on the target Windows system.

When SMB signing is disabled, communication between client and server is not digitally verified. This allows an attacker to perform:

- Man-in-the-Middle (MITM) attacks
- Session hijacking
- SMB relay attacks
- Data tampering during transmission

An unauthenticated remote attacker could intercept or modify SMB traffic.

Root Cause

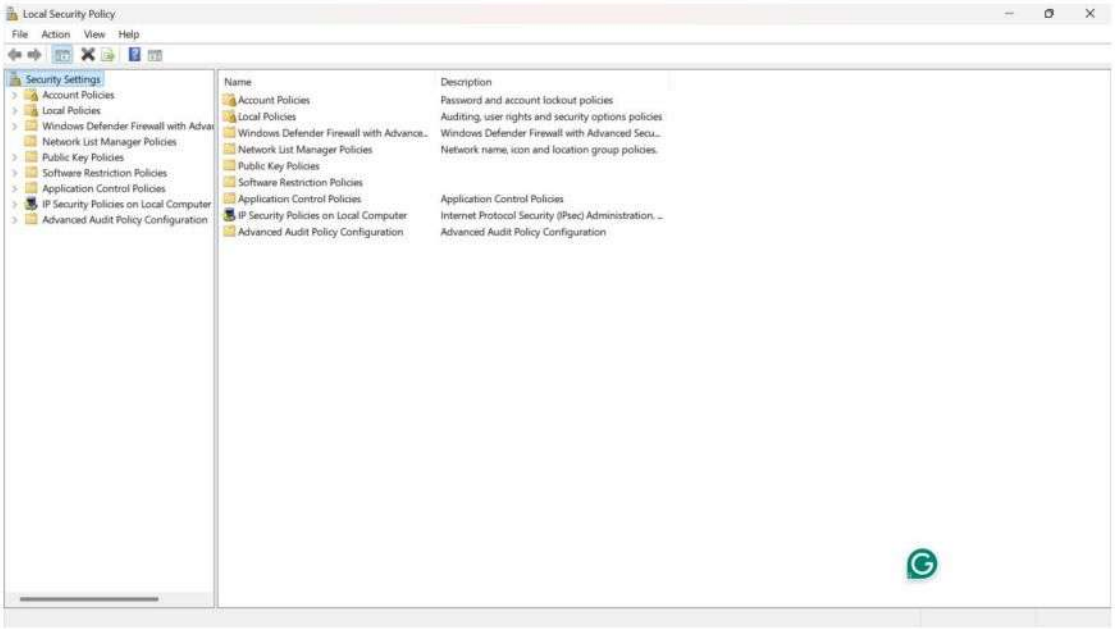
The following security policy was not enforced:

> **Microsoft network server: Digitally sign communications (always)**

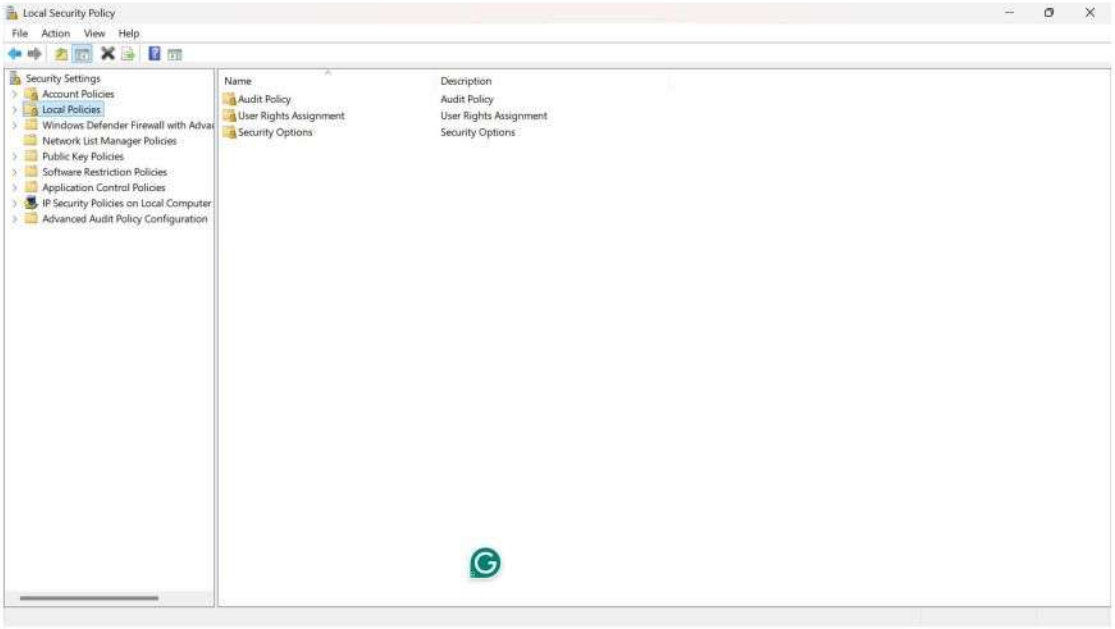
Because it was not mandatory, SMB communication could occur without digital signing.

STEP-BY-STEP REMEDIATION PERFORMED

Step 1: Open Local Security Policy

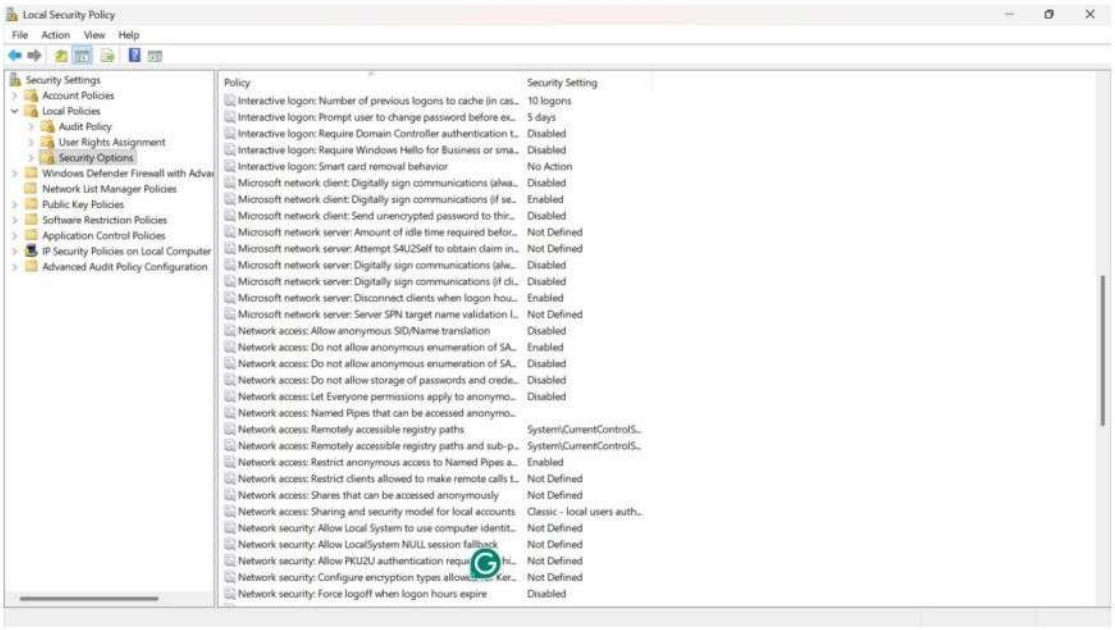


Step 2: Navigate to SMB Signing Setting



Locate:

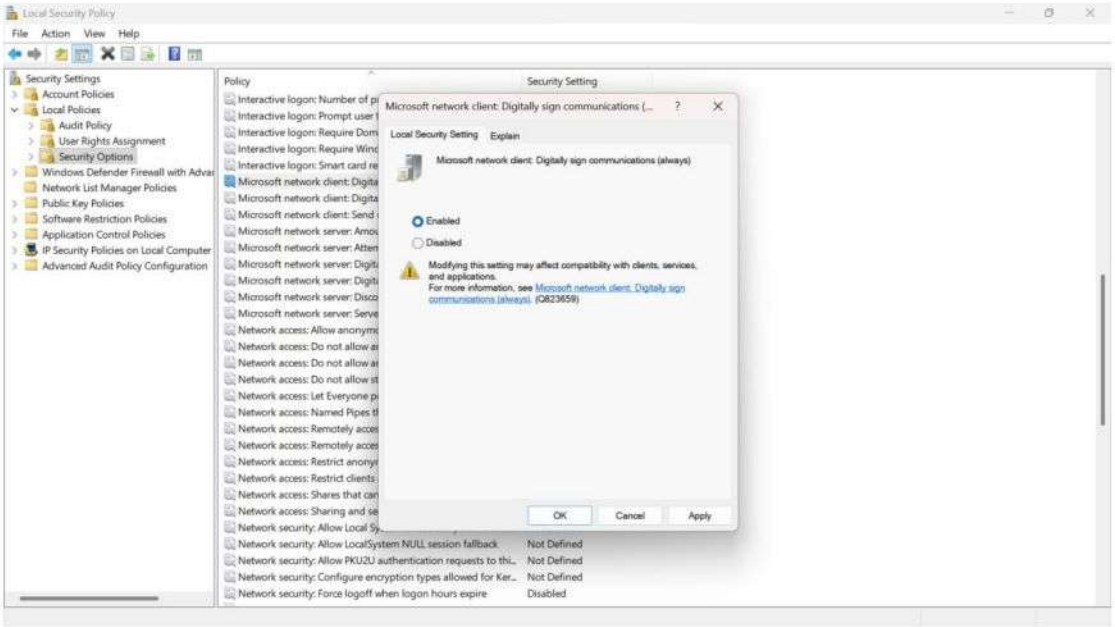
Microsoft network server: Digitally sign communications (always)

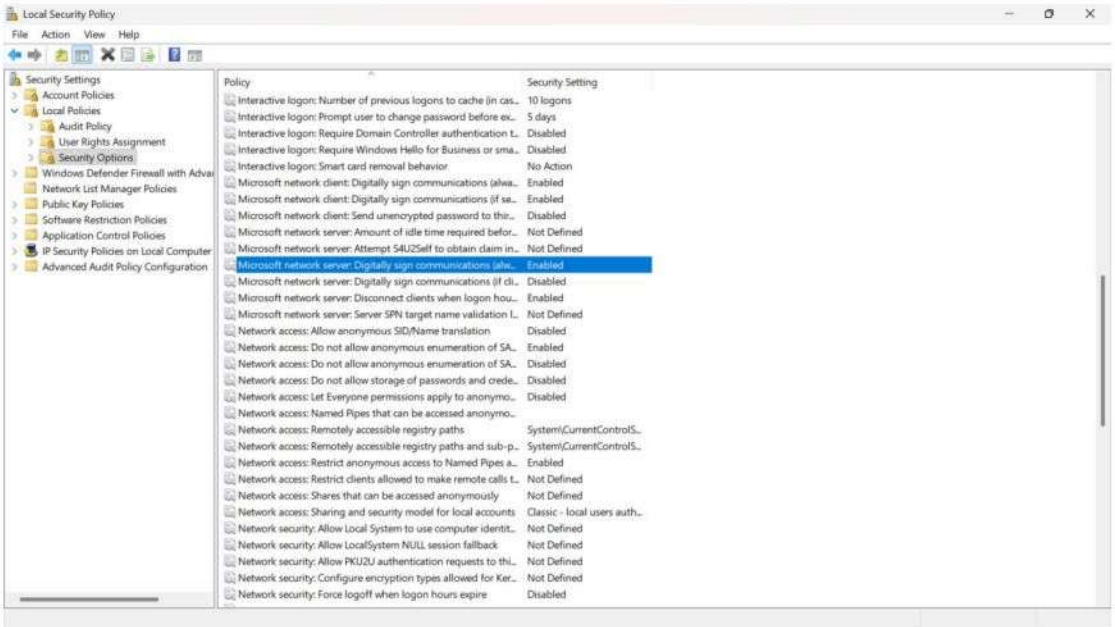


[found 2]

Step 3: Enable the Policy

1. Double-click the setting
2. Select: **Enabled**
3. Click **Apply**
4. Click **OK**

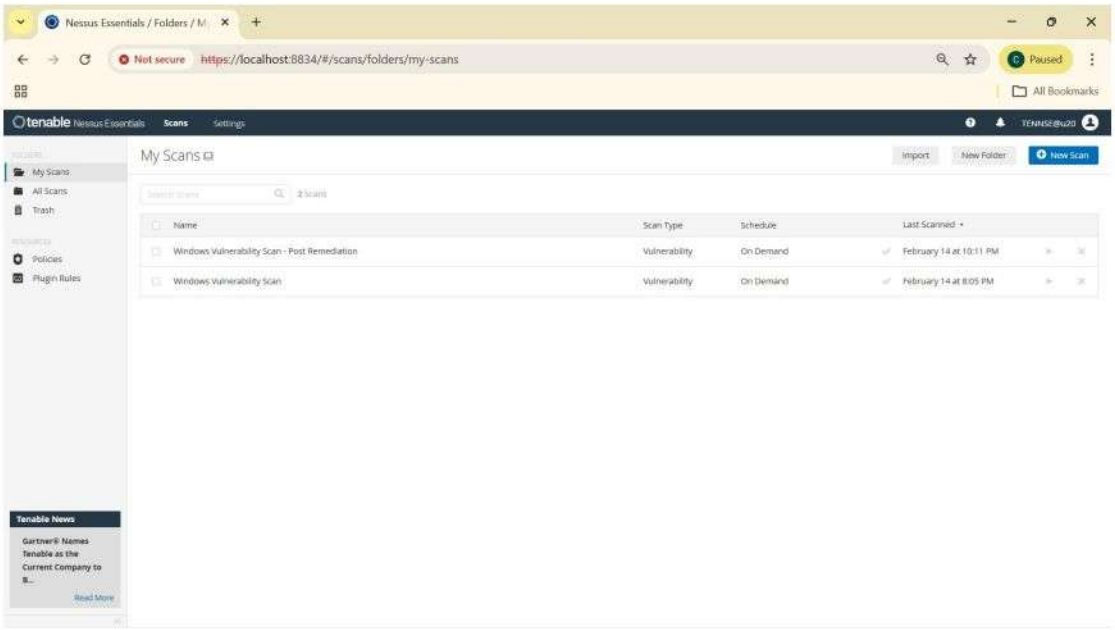


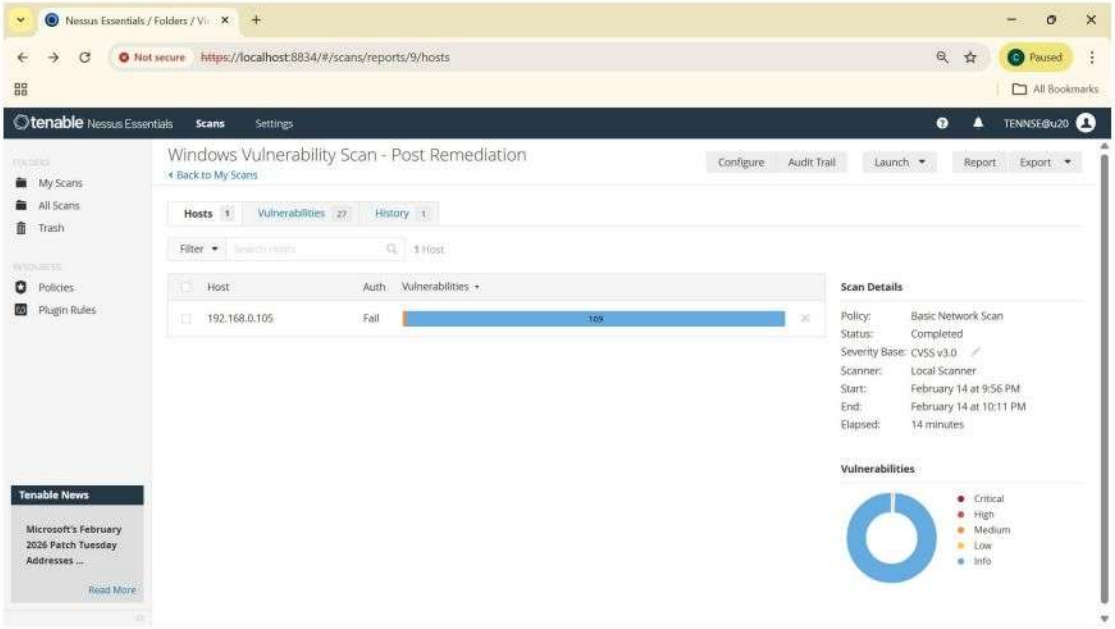


Step 4: Update and Restart the System

POST REMEDIATION SCAN RESULT

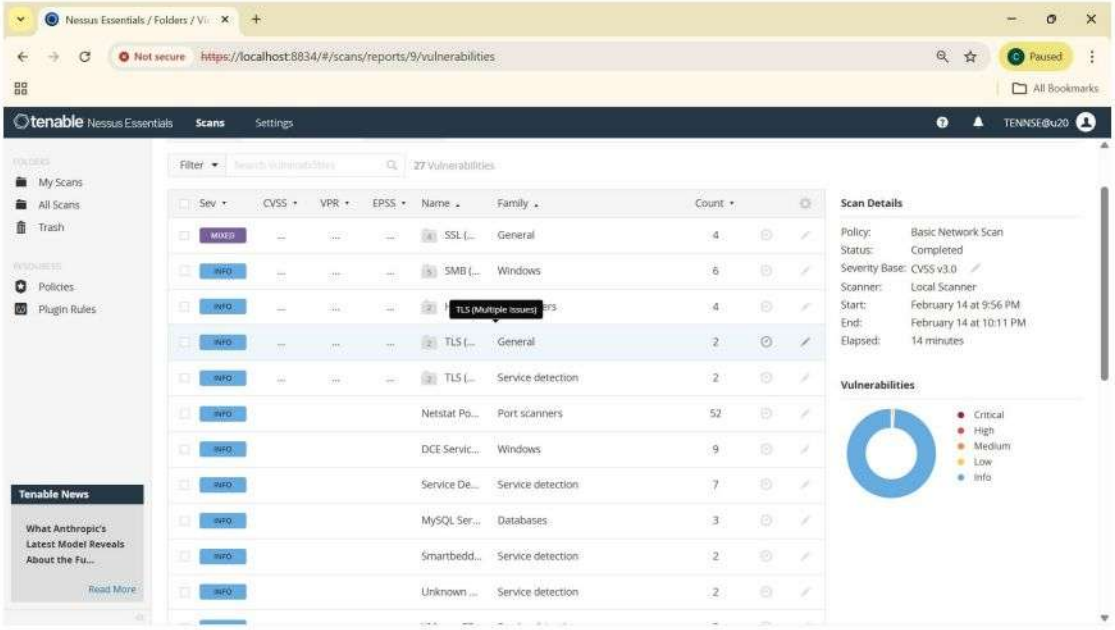
After implementing recommended security configurations (SMB signing enforcement and policy adjustments), a second scan was performed.





Results:

- Initial Scan Vulnerabilities: 28
- Post Remediation Vulnerabilities: 27
- Severity: No Critical or High vulnerabilities detected
- Majority findings: Informational



Observation:

The reduction in vulnerability count indicates that remediation measures were partially effective. However, informational findings related to open ports and service enumeration remain.

Conclusion of Remediation:

The system security posture improved after applying configuration changes, but further hardening (firewall restrictions and service minimization) is recommended.

RISK CLASSIFICATION AND PRIORITIZATION

The vulnerabilities discovered during the assessment were classified based on CVSS v3.0 scoring provided by Tenable Nessus.

Priority Decision:

- The Medium severity vulnerability (SMB Signing Not Required) was treated as **Priority 1** and remediated immediately.
- Informational vulnerabilities are considered low risk but require monitoring and system hardening.

PREVENTION AND HARDENING STEPS

To reduce attack surface and improve overall security posture, the following hardening measures are recommended:

- Enable SMB message signing (Completed)
- Keep Windows OS updated with latest security patches
- Disable unused services and ports
- Restrict RPC and NetBIOS exposure via firewall rules
- Enable Windows Firewall after scanning
- Implement strong password policies
- Regular vulnerability scanning schedule (monthly)

BUSINESS IMPACT ANALYSIS

The vulnerabilities identified during the vulnerability assessment could have significant security implications if left unaddressed.

The Medium severity vulnerability **"SMB Signing Not Required"** could allow an attacker to perform Man-in-the-Middle (MITM) attacks. This may lead to:

- Unauthorized interception of network traffic
- Credential theft
- Data manipulation
- Loss of data confidentiality and integrity

Even though the majority of findings were Informational, exposed services such as SMB, RPC, NetBIOS, and open ports increase the system's attack surface. If exploited in a real-world environment, this could result in:

- System compromise
- Unauthorized remote access
- Data leakage
- Service disruption

By remediating the identified vulnerability and implementing recommended hardening measures, the overall security posture of the Windows system has been significantly improved.

CONCLUSION

The Vulnerability Assessment of the Windows Operating System was successfully conducted using **Tenable Nessus Essentials**. The scan identified multiple security findings, including one Medium severity vulnerability and several Informational findings related to open ports, service detection, and system configurations.

The Medium severity vulnerability **"SMB Signing Not Required"** was analyzed in detail and successfully remediated by enforcing message signing through Local Security Policy settings. A post-remediation scan confirmed the improvement in the system's security posture.

Although most of the findings were categorized as Informational, they provided valuable insight into exposed services and potential attack surfaces. Proper hardening measures such as enabling SMB signing, restricting unnecessary services, keeping the system updated, and enforcing firewall rules were implemented to strengthen system security.

This project demonstrates the importance of vulnerability assessment in identifying security weaknesses before they can be exploited. Regular scanning, proper risk prioritization, and timely remediation are essential practices to maintain a secure computing environment.

Overall, the Windows system is now more secure compared to its initial scan state, and this assessment highlights the effectiveness of proactive security monitoring and remediation practices.